

Lesson 14

Quadratic Functions & the Discriminant

Functions · 60-minute lesson

Syllabus SL 2.6–2.7 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Factorise $x^2 - 5x + 6$.
2. Solve $x^2 - 9 = 0$.
3. What shape is the graph of $y = x^2$?

Answers (click to reveal)

1. $(x-2)(x-3)$
2. $x = \pm 3$
3. a parabola (U-shaped)

What we're learning today

LEARNING OBJECTIVES

- 1** **Recognise**
the forms of a quadratic and what each tells you.
- 2** **Use**
the quadratic formula to solve.
- 3** **Apply**
the discriminant $\Delta = b^2 - 4ac$.
- 4** **Determine**
the number of real roots from Δ .

SUCCESS CRITERIA — I CAN...

- ✓
read the y-intercept and shape from ax^2+bx+c
- ✓
solve a quadratic with the formula
- ✓
calculate the discriminant
- ✓
decide how many real roots a quadratic has

Solving quadratics

THE FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a, b, c

the coefficients of x^2 , x and the constant

$\pm$ gives

up to two solutions

OTHER FORMS

Factorised

$y = a(x - p)(x - q)$: roots p, q

Vertex

$y = a(x - h)^2 + k$: vertex (h, k)

Standard

$y = ax^2 + bx + c$: c is the y -intercept

Solve with the formula

QUESTION

Solve $x^2 - 4x + 3 = 0$
using the quadratic formula.

SOLUTION

$a = 1$, $b = -4$, $c = 3$. Substitute:

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(3)}}{2}$$

Simplify the discriminant ($16 - 12 = 4$):

$$x = \frac{4 \pm 2}{2}$$

Two solutions:

$$x=3 \quad \vee \quad x=1$$

Your turn — solve

ANSWER IN YOUR BOOK

1. Solve $x^2 - 7x + 10 = 0$.

2. Solve $x^2 + 2x - 8 = 0$.

3. Solve $x^2 - 6x + 9 = 0$.

Answers (click to reveal)

1. $x = 5$ or 2

2. $x = 2$ or -4

3. $x = 3$ (repeated)

The discriminant

$$\Delta = b^2 - 4ac$$

$$= b^2 - 4ac$$

$$\Delta > 0$$

two distinct real roots

$$\Delta = 0$$

one repeated real root

$$\Delta < 0$$

no real roots

WHY IT WORKS

Δ is what's under the root in the formula.

Positive

$\sqrt{\quad}$ gives two values.

Zero

the $\pm$ does nothing $\rightarrow$ one root.

Negative

can't $\sqrt{\quad}$ a negative (no real roots).

Use the discriminant

QUESTION

How many real roots does

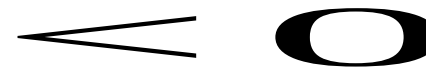
$$x^2 + 2x + 5 = 0 \text{ have?}$$

SOLUTION

$a = 1$, $b = 2$, $c = 5$. Compute Δ :

$$= 2^2 - 4(1)(5) = 4 - 20$$

$\Delta = -16$, which is negative:



So there are NO real roots.

Exam-style question

The equation $x^2 + kx + 9 = 0$ has equal roots, $k > 0$.

- (a) Write down the condition on the discriminant. [1]
- (b) Find the value of k . [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $\Delta = 0$ (A1)
- (b) $k^2 - 4(1)(9) = 0$ (M1) $\rightarrow k^2 = 36$ (A1) $\rightarrow k = 6$ ($k > 0$) (A1)

Common mistakes



Sign of b

In the formula it is $-b$; watch negatives carefully.



Δ meaning

Δ tells the NUMBER of roots, not the roots themselves.



Equal roots

'Equal/repeated roots' means $\Delta = 0$.



y-intercept

In ax^2+bx+c the y-intercept is c , found at $x = 0$.

Mixed practice

QUESTIONS

1. Solve $x^2 - 2x - 15 = 0$.
2. Find Δ for $2x^2 + 3x + 1$.
3. How many roots has $x^2 + x + 4 = 0$?
4. For $x^2 - 6x + k = 0$ with equal roots, find k .

Answers (click to reveal)

1. $x = 5$ or -3
2. $9 - 8 = 1$ (two roots)
3. $\Delta = 1 - 16 < 0 \rightarrow$ none
4. $36 - 4k = 0 \rightarrow k = 9$

Plenary & exit ticket

KEY FORMULAS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

EXIT TICKET — before you leave

1. Solve $x^2 - x - 6 = 0$.
2. Find Δ for $x^2 + 4x + 4$.
3. How many roots if $\Delta = -5$?

Answers (click to reveal)

1. $x = 3$ or -2
2. 0 (one repeated root)
3. none

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Quadratics & the discriminant.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 15

Rational & reciprocal functions — graphs with asymptotes.

$$y = \frac{ax + b}{cx + d}$$